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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

CC4221-1 –INTRODUCTION TO BLOCKCHAIN TECHNOLOGY
(For CC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **Even if a group of nodes fails, the distributed system continues to operate correctly and what is it called?**
A) Partition tolerance
B) Confidentiality
C) Integrity
D) Availability
2. **Which property ensures that all nodes in a distributed system have a single latest copy of data?**
A) Consistency
B) Transparency
C) Inconsistency
D) Immutability
3. **What is a key benefit of disintermediation in blockchain technology?**
A) Faster transactions with lower costs due to the removal of intermediaries.
B) Higher dependency on banks and third-party services.
C) Increased risk of manipulation by a central authority.
D) Longer processing times due to regulatory oversight.
4. **Which blockchain consensus mechanism relies on competition to validate transactions?**
A) Proof of Work (PoW), where miners compete to solve cryptographic puzzles.
B) Proof of Stake (PoS), where validators are chosen randomly.
C) Federated Byzantine Agreement, where only pre-selected nodes validate transactions.
D) Proof of Importance (PoI), where transactions are prioritized based on social contribution.
5. **Which of the following is an example of symmetric encryption?**
A) RSA
B) AES
C) Diffie-Hellman
D) Elliptic Curve Cryptography (ECC)
6. **Which of the following is an example of a stream cipher?**
A) DES
B) AES
C) RC4
D) SHA-256
7. **Which of the following is NOT a block cipher?**
A) AES
B) DES
C) Blowfish
D) RC4
8. **A keystream generator is commonly used in:**
A) Block ciphers
B) Stream ciphers
C) Hash functions
D) Digital signatures
9. **Smart contracts eliminate the need for:**
A) Cryptography
B) A centralized authority or intermediaries
C) Digital signatures
D) Internet connectivity

10. **What is a Ricardian Contract?**
 A) A smart contract with legal and machine-readable components
 B) A simple agreement between two users
 C) A contract that does not require legal approval
 D) A traditional contract stored on paper
11. **The native cryptocurrency of Ethereum is:**
 A) Bitcoin (BTC)
 B) Litecoin (LTC)
 C) Ether (ETH)
 D) Ripple (XRP)
12. **Ethereum was launched in:**
 A) 2008
 B) 2013
 C) 2015
 D) 2017
13. **What versioning system is used in Solidity files?**
 A) SPDX-License-Identifier
 B) JavaScript Import System
 C) Solidity Blockchain Versioning
 D) Ethereum Virtual Machine (EVM) Identifier
14. **What are the basic value types in Solidity?**
 A) Boolean, Integers, Address, Bytes
 B) Structs, Arrays, Enums, Contracts
 C) Strings, Tuples, Objects, Classes
 D) Doubles, Floats, Characters, Lists
15. **What is the default value of an uninitialized integer in Solidity?**
 A) 0
 B) 1
 C) null
 D) Undefined
16. **Address type in Solidity is used to store:**
 A) IP addresses
 B) Email addresses
 C) Ethereum wallet addresses
 D) URL links
17. **Which of the following is a key component of Hyperledger's governance structure?**
 A) A single controlling authority
 B) Community-driven governance
 C) Private software updates
 D) Closed-source development
18. **Which organization contributed Hyperledger Fabric?**
 A) Intel
 B) IBM
 C) Ripple
 D) Google
19. **What is Hyperledger Explorer used for?**
 A) Exploring uncharted blockchain networks
 B) Viewing and analyzing blockchain data
 C) Mining cryptocurrency
 D) Validating smart contracts
20. **Hyperledger Fabric supports:**
 A) Only public blockchain networks
 B) Only permissioned networks
 C) Both permissioned and permissionless networks
 D) Only private organizations

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) List and explain the generic elements of a blockchain.	10	L*2	1	1,2
	b) Explain the benefits and limitations of a blockchain.	6	L2	1	1,2
2.	a) Briefly explain the following ideas in the context of blockchain:				
	i. Consensus mechanisms				
	ii. Fault tolerance-based mechanism				
	iii. Distributed systems				
	iv. Semi-private blockchain				
	v. Proof of importance	10	L2	1	1,2
	b) Explain how any system can be decentralized with an example.	6	L2	1	1,2
3.	a) Discuss the most common block encryption modes.	10	L2	2	1,2
	b) Write the differences between stream and block ciphers.	6	L2	2	1,2

Unit – II

4. a)	Define a Ricardian contract and explain its purpose. What is the Bowtie Model in Ricardian contracts? List at least five key properties of a Ricardian contract. Differentiate between operational semantics and denotational semantics in contracts.	12	L2	3	1,2
b)	Explain the key differences between Ricardian contracts and smart contracts.	4	L2	3	1,2
5. a)	Explain the structure of Ethereum stack with a neat diagram.	6	L2	3	1,2
b)	Explain the structure of an Ethereum transaction and the purpose of its main fields.	10	L2	3	1,2
6. a)	Implement a smart contract called 'SimpleMapping' that uses a mapping to store and retrieve ages associated with names.	8	L3	4	2
b)	Implement a smart contract called 'Calculator' that performs the following operations: • Reads and returns the current value of a state variable count using a view function. • Calculates the square of a given number without modifying the contract's state using a pure function. • Increments the count state variable, calculates the square of a given number using the pure function, and retrieves the updated count value using the view function.	8	L3	4	2

Unit – III

7. a)	Discuss the building blocks of blockchain solutions.	8	L2	5	2
b)	What are the key tools provided by Hyperledger, and what are their functions? Explain.	8	L2	5	2
8. a)	What are the fundamental components that make up a blockchain solution? Explain in detail.	8	L2	5	2
b)	What are the key principles that guide Hyperledger's design philosophy? Explain in detail.	8	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
